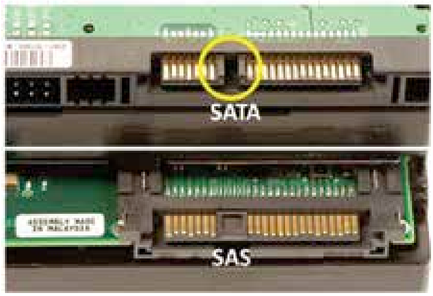
however, as time passes the fragmentation of data lead to much slower speeds. PATA needed so many connections since it would transfer data parallelly, i.e. each data line had one bit moving along it simultaneously. PATA was also known by the name IDE(Integrated Drive Electronics) or EIDE(Enhanced Integrated Drive Electronics).

SATA

SATA on the other hand uses serial mode of data transfer, so a stream of data would be sent in one burst leading to much higher speeds. SATA has undergone quite a few revisions and is now in it's third avatar. The speed has also kept with the times, SATA had an upper limit of 150 MB/s and SATA2 doubled that with 300MB/s. SATA3 doubled even that to hit 600 MB/s. Hard drives could barely sustain 110 MB/s on their own so one wonders why was the new interface needed in the first place. The answer lies in RAID systems, this allowed people to utilise more than one hard drive as a single drive. Thus each drive transferring at 110 MB/s would contribute and cumulative the entire system would have increased transfer speed. Nowadays you can easily hit the SATA3 limit with two SSDs in RAID 0 configuration.



SCSI / SAS

SCSI or Small Computer System Interface is a technology attachment that is catered towards enterprise and server solutions. They are more optimised for systems where more than one hard drive is to be installed in single, JBOD(Just a Bunch of Drives) or RAID(Redundant Array of Independent Disks) configurations. They usually are faster, have low seek times due to high RPM and have hardware monitoring for the hard drive's health. A variant of this is the SAS (Serial Attached SCSI). This interface ushered in even greater throughput than before. The connector is the same as SATA so normal desktop users can switch to SAS drives for the extra speeds. However, with the advent of SSDs and hybrid drives, this seems to be a more expensive option.